2D Matrices: Eigenvalues and Eigenvectors

Questions 1–11

Question 1. Show that each of the following vectors are eigenvectors of the given matrix $\mathbf{A}$ and in each case state the associated eigenvalue:

a) $\begin{pmatrix} 2 \\ -1 \end{pmatrix}, \mathbf{A} = \begin{pmatrix} 4 & 2 \\ 1 & 5 \end{pmatrix}$

b) $\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \mathbf{A} = \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}$

c) $\begin{pmatrix} 1 \\ -2 \end{pmatrix}, \mathbf{A} = \begin{pmatrix} 0 & 1 \\ 2 & -1 \end{pmatrix}$

Question 2. It is given that $\mathbf{M} = \begin{pmatrix} 3 & -5 \\ 0 & -2 \end{pmatrix}$

a) Show that the characteristic equation of $\mathbf{M}$ is given by $\lambda^2 - \lambda - 6 = 0$ where λ is an eigenvalue of $\mathbf{M}$

b) Hence state the eigenvalues of $\mathbf{M}$

Question 3. Find the eigenvalues of each of the following matrices:

a) $\begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$

b) $\begin{pmatrix} -3 & 6 \\ -2 & 1 \end{pmatrix}$

c) $\begin{pmatrix} 0 & -5 \\ 1 & 0 \end{pmatrix}$

d) $\begin{pmatrix} 2 & 7 \\ 0 & 2 \end{pmatrix}$

e) $\begin{pmatrix} -1 & 4 \\ -3 & 5 \end{pmatrix}$

f) $\begin{pmatrix} 6 & -1 \\ 2 & 3 \end{pmatrix}$

g) $\begin{pmatrix} \frac{1}{2} & -\frac{3}{4} \\ \frac{3}{4} & \frac{1}{2} \end{pmatrix}$

Question 4. It is given that $\mathbf{M} = \begin{pmatrix} 5 & 2 \\ 1 & 4 \end{pmatrix}$

a) Show that the characteristic equation of $\mathbf{M}$ is given by $\lambda^2 - 9\lambda + 18 = 0$ where λ is an eigenvalue of $\mathbf{M}$

b) Hence state the eigenvalues and corresponding eigenvectors of $\mathbf{M}$

Question 5. Determine the eigenvalues and eigenvectors for each of the following:

a) $\begin{pmatrix} 4 & 2 \\ -3 & -3 \end{pmatrix}$

b) $\begin{pmatrix} -87 & 120 \\ -64 & 89 \end{pmatrix}$

c) $\begin{pmatrix} 60 & 10 \\ -10 & 40 \end{pmatrix}$

d) $\begin{pmatrix} 67 & -54 \\ 27 & -32 \end{pmatrix}$

e) $\begin{pmatrix} \frac{4}{15} & \frac{2}{3} \\ \frac{1}{3} & -\frac{1}{15} \end{pmatrix}$

f) $\begin{pmatrix} -\frac{26}{15} & \frac{4}{5} \\ \frac{4}{5} & \frac{2}{5} \end{pmatrix}$

Question 6. Given that the matrix $\mathbf{M} = \begin{pmatrix} 6 & -2 \\ k-3 & 2 \end{pmatrix}$ has a repeated eigenvalue, find the value of k

Question 7. Given that in each case, the matrix $\mathbf{M}$ has two distinct real eigenvalues, find the range of values for k where k is a real constant:

a) $\mathbf{M} = \begin{pmatrix} 4 & 1 \\ k & 3 \end{pmatrix}$

b) $\mathbf{M} = \begin{pmatrix} k & 2 \\ 3 & 5 \end{pmatrix}$

Question 8. The matrix $\mathbf{M} = \begin{pmatrix} -1 & k \\ 4 & k-1 \end{pmatrix}$ where k is a real constant.

a) Show that the characteristic equation of $\mathbf{M}$ is given by $\lambda^2 + (2-k)\lambda - (5k-1) = 0$ where λ is an eigenvalue of $\mathbf{M}$

b) Find the range of values for k for which $\mathbf{M}$ has two distinct real eigenvalues.

c) Given that $k = 9$, find the eigenvalues and corresponding eigenvectors of $\mathbf{M}$ giving your answer in terms of k

Question 9. Show that the matrix

a) $\begin{pmatrix} p & -q \\ q & p \end{pmatrix}$ for constants p and q where $q \neq 0$ has complex eigenvalues $\lambda = p \pm qi$

b) $\begin{pmatrix} p & q \\ 0 & p \end{pmatrix}$ for constants p and q where $q \neq 0$ has eigenvalue given by $\lambda = p$

Question 10. Prove that if λ is an eigenvalue of the matrix $\mathbf{M}$, then λ^n is an eigenvalue of $\mathbf{M}^n$ where n is a positive whole number.

Question 11. For each of the following, $\mathbf{v}$ is an eigenvector of both the matrix $\mathbf{A}$ and matrix $\mathbf{B}$. The corresponding eigenvalues for $\mathbf{v}$ in each case are also given, with an equation involving matrix $\mathbf{A}$, matrix $\mathbf{B}$, $\mathbf{v}$ and λ. Find the value of λ

a) $\lambda_A = 3, \lambda_B = 2, \mathbf{AB}\mathbf{v} = \lambda\mathbf{v}$
b) $\lambda_A = 4, \lambda_B = 5, \mathbf{A}^2\mathbf{B}\mathbf{v} = \lambda\mathbf{v}$
c) $\lambda_A = -1, \lambda_B = 2, \mathbf{A}\mathbf{B}^3\mathbf{v} = \lambda\mathbf{v}$
d) $\lambda_A = -2, \lambda_B = -4, \mathbf{A}^2\mathbf{B}^3\mathbf{v} = \lambda\mathbf{v}$
e) $\lambda_A = p, \lambda_B = q, \mathbf{A}^m\mathbf{B}^n\mathbf{v} = \lambda\mathbf{v}$ for constants p and q and positive constants m and n